Algebra Monsoon 2026 Lecture 17 Admissible filtrations, multiplicities

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Admissible filtrations

Let G be a finite group.

A filtration on G is a decreasing family of subgroups $(G_i)_{i \geq 0}$ such that

1) $G_0 = G$ and $G_i = \{e\}$ for $i \gg 0$

2) G_{i+1} is normal in G_i for all $i \geq 0$.

A filtration said to be admissible if G_i / G_{i+1} is simple or trivial for all $i \geq 0$.

- Let $(G_i)_{i \geq 0}$ be a filtration on G. For a subgroup $H \subset G$, put $H_i = G_i \cap H$. Then $(H_i)_{i \geq 0}$ is a filtration on H; it is admissible if $(G_i)_{i \geq 0}$ is admissible.

- Let $(G_i)_{i\geq 0}$ be a filtration on G. For every quotient $\pi : G \twoheadrightarrow Q$, put $Q_i = \pi_*(G_i)$. Then $(Q_i)_{i\geq 0}$ is a filtration on Q; it is admissible if $(G_i)_{i\geq 0}$ is admissible.

Lemma Every finite group admits an admissible filtration.

Proof: By induction on $o(G)$. If $G = \{e\}$ or if G is simple, this is clear.

Suppose $o(G) > 1$ and G has a normal subgroup $N \neq \{e\}, G$. By induction hypothesis, N and $Q = G/N$ have admissible filtrations $(N_i)_{i\geq 0}$, $(Q_j)_{j\geq 0}$.

There is an n such that $Q_j = \{e\}$ for $j \geq n$. Let $\pi : G \to Q$ be the canonical surjection, and consider the filtration

$$\cdots \subset N_2 \subset N_1 \subset N = \pi^*(Q_n) \subset \cdots \subset \pi^*(Q_1) \subset \pi^*(Q_0)$$

It can be checked that this is an admissible filtration on G. □

Proposition Let $N \subset G$ be a normal subgroup, put $Q = G/N$ and let $\pi : G \to Q$ be the canonical surjection. For every admissible filtration $(G_i)_{i \geq 0}$ on G, put $N_i = N \cap G_i$ and $Q_i = \pi_*(G_i)$. We claim that $(N_i)_{i \geq 0}$ and $(Q_i)_{i \geq 0}$ are admissible filtrations on N and Q, and the sequence

$$\{e\} \to N_i/N_{i+1} \to G_i/G_{i+1} \to Q_i/Q_{i+1} \to \{e\}$$

is exact.

Proof $(N_i)_{i \geq 0}$ is admissible :

- $N_0 = N$ because $G_0 = G$.
- $N_i = \{e\}$ for $i >> 0$ because $G_i = \{e\}$.
- the induced mapping

$$\begin{array}{ccccc} N_i & \longrightarrow & G_i & \longrightarrow & G_i/G_{i+1} \\ \downarrow & & & \nearrow & \\ N_i/N_{i+1} & & & & \end{array}$$

is injective, so N_i/N_{i+1} is trivial or

simple.

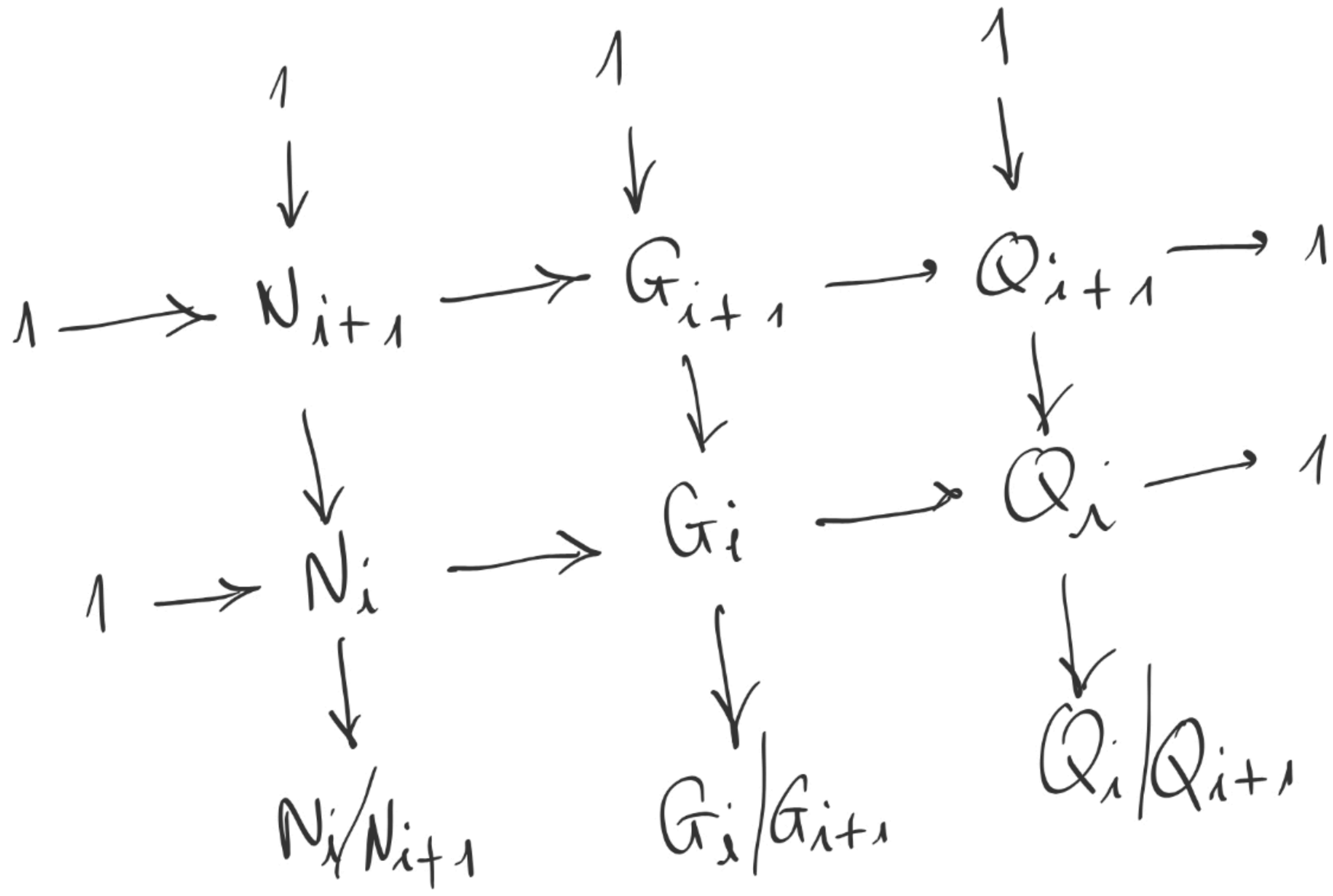
The filtration $(Q_i)_{i\geq 0}$ is admissible:

- $Q_0 = Q$ because $G_0 = G$
- $Q_i = \{e\}$ for $i >> 0$ because $G_i = \{e\}$.
- $Q_{i+1} \subset Q_i$ is the image of a normal subgroup $G_{i+1} \subset G_i$ by a surjective morphism $G_i \longrightarrow Q_i$, so Q_{i+1} is normal in Q_i. As the induced morphism

$$G_i/G_{i+1} \longrightarrow Q_i/Q_{i+1}$$

is surjective, we conclude that Q_i/Q_{i+1} is trivial or simple.

Applying the snake lemma to the diagram



yields the exact sequence in question :

$$1 \to N_i/N_{i+1} \to G_i/G_{i+1} \to Q_i/Q_{i+1} \to 1.$$

□

Let G be a finite group, $(G_i)_{i \geq 0}$ an admissible filtration on G, and let S be a finite simple group.

Def $n(G, (G_i)_{i\geq 0}, S)$ = the number of indices i such that G_i/G_{i+1} is isomorphic to S.

$$n(\mathbb{Z}/p\mathbb{Z}, (\mathbb{Z}/p\mathbb{Z})_{i\geq 0}, S) = 1.$$

Example. $n(S, (S_i)_{i\geq 0})$ for any admissible filtration $(S_i)_{i\geq 0}$.

- If T is a finite simple group not isomorphic to S, then $n(T, (T_i)_{i\geq 0}, S) = 0$ for any admissible filtration $(T_i)_{i\geq 0}$.

Theorem (Jordan-Hölder) $n(G, (G_i)_{i\geq 0}, S)$ depends only on G and S, not on the admissible filtration $(G_i)_{i\geq 0}$.

Proof By induction on $o(G)$. This is clear if G is trivial or simple. Suppose that G has a normal subgroup $N \neq \{e\}, G$, let $(G_i)_{i\geq 0}$ be an admissible filtration on G, and let $(N_i)_{i\geq 0}, (Q_i)_{i\geq 0}$ be the induced filtrations on N and $Q = G/N$ respectively; they are both admissible We have the exact sequence

$$1 \longrightarrow N_i/N_{i+1} \longrightarrow G_i/G_{i+1} \longrightarrow Q_i/Q_{i+1} \longrightarrow 0$$

so $m(G,(G_i)_{i\geq 0},S) = m(N,(N_i),S) + m(Q,(Q_i)_{i\geq 0},S)$.

As the RHS is independent of the admissible filtrations by the induction hypothesis, so is the left hand side. □

Corollary Let S be a finite simple group. There is a unique function

$$m_S : (\text{finite groups}) \longrightarrow \mathbb{N}$$

such that (1) $m_S(G) = 1 \iff G$ is isomorphic to S

and (2) $m_S(G) = m_S(G') + m_S(G'')$ whenever there is an exact sequence

$$1 \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow 1.$$

Proof Indeed, necessarily

$$m_S(G) = m(G,(G_i)_{i\geq 0},S)$$

$m_S(G) = n(G, (G_i)_{i\geq 0})$

for any admissible filtration $(G_i)_{i\geq 0}$ on G. □

We have highlighted the analogy between

$$v_p : \mathbb{N}^* \longrightarrow \mathbb{N}$$

and $m_S : \underbrace{\mathcal{G}_0}_{\text{the category of finite groups.}} \longrightarrow \mathbb{N}$

There is one major difference, however:

Every $n \in \mathbb{N}^*$ is determined by the family $\left(v_p(n)\right)_p$:

$$n = \prod_p p^{v_p(n)}.$$

However, G is *not* determined by the family $\left(m_S(G)\right)_S$ in general:

we have $m_S(G_3) = m_S(C_6)$

for every S, and yet C_3 is not isomorphic to C_6 because C_6 is commutative and C_3 is not. Even simpler $m_S(C_2 \times C_2) = m_S(C_4)$ for every simple group S.

In fact, Bourbaki derives the uniqueness of prime factorisation of $a \in \mathbb{N}^*$ by applying the Jordan-Hölder theorem to

$$G = \mathbb{Z}/a\mathbb{Z} :$$

if $a = p_1^{r_1} \cdots p_n^{r_n}$

then $\mathbb{Z}/a\mathbb{Z}$ has the admissible filtration coming from

$$\cdots \subset p_1^{r_1} p_2 \mathbb{Z} \subset p_1^{r_1} \mathbb{Z} \subset \cdots \subset p_1^2 \mathbb{Z} \subset p_1 \mathbb{Z} \subset \mathbb{Z}$$

so $m_{\mathbb{Z}/p\mathbb{Z}}(G) = r_1$.